

SDS 313 HW

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4/14/25

<https://github.com/SamuelLi1225/SDS-HW8>

```
##
## Call:
## lm(formula = creatclear ~ age, data = creatinine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.2249  -4.6175   0.2221   4.7212  15.8221
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 147.81292    1.37965  107.14  <2e-16 ***
## age         -0.61982    0.03475  -17.84  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.911 on 155 degrees of freedom
## Multiple R-squared:  0.6724, Adjusted R-squared:  0.6703
## F-statistic: 318.2 on 1 and 155 DF,  p-value: < 2.2e-16
## [1] 113.7228
##
## Call:
## lm(formula = creatclear ~ age, data = creatinine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.2249  -4.6175   0.2221   4.7212  15.8221
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 147.81292    1.37965  107.14  <2e-16 ***
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##
##      1
## 11.97972
```

```
##          1
## 1.376035

#1.A

From the model, the y intercept was 147.81292 and the slope was -0.61982, so i subtracted the intercept by
55 times the slope to predict the levels for 55 year olds. My prediction was 113.7228

#1.B

The model states that the slope is -0.61892. Creatinine clearance decreases by about 0.62 ml/min per year.

#1.C

The 40 year old has a residual of 11.97972 and the 60 year old has a residual of 1.376035. This means that
the 40 year old is predicted to be healthier as he has a higher residual.

## Registered S3 method overwritten by 'mosaic':
##   method                from
##   fortify.SpatialPolygonsDataFrame ggplot2

##
## The 'mosaic' package masks several functions from core packages in order to add
## additional features. The original behavior of these functions should not be affected by this.

##
## Attaching package: 'mosaic'

## The following objects are masked from 'package:dplyr':
##
##   count, do, tally

## The following object is masked from 'package:Matrix':
##
##   mean

## The following object is masked from 'package:ggplot2':
##
##   stat

## The following objects are masked from 'package:stats':
##
##   binom.test, cor, cor.test, cov, fivenum, IQR, median, prop.test,
##   quantile, sd, t.test, var

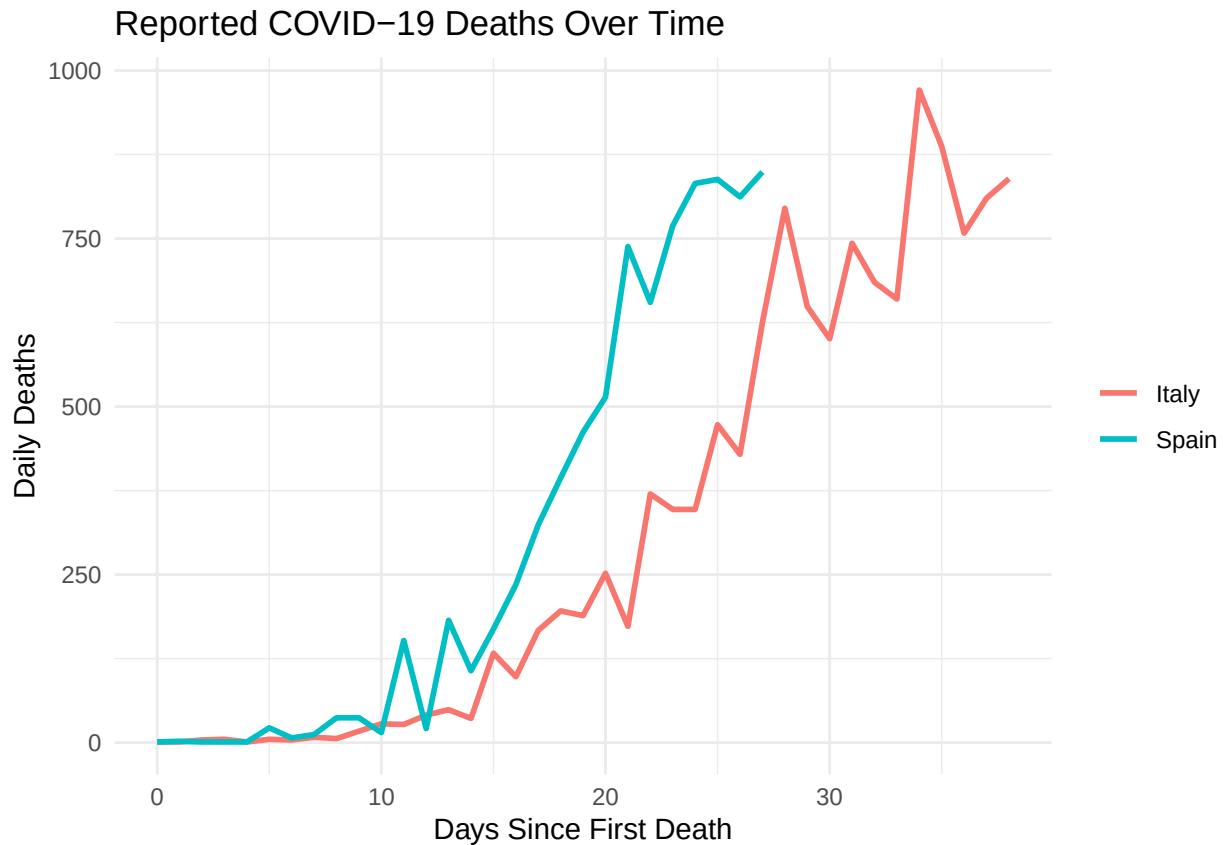
## The following objects are masked from 'package:base':
##
##   max, mean, min, prod, range, sample, sum

##
## Call:
## lm(formula = log_deaths ~ days_since_first_death, data = italy)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.7515 -0.4711  0.1435  0.5130  1.1235
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.01860    0.22773   4.473 7.09e-05 ***
## days_since_first_death 0.18322    0.01031  17.766 < 2e-16 ***
```

```

## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7248 on 37 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8922
## F-statistic: 315.6 on 1 and 37 DF,  p-value: < 2.2e-16
##
## Call:
## lm(formula = log_deaths ~ days_since_first_death, data = spain)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.57020 -0.54979  0.01883  0.54766  1.51997
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.46522    0.30066   1.547   0.134
## days_since_first_death 0.27624    0.01911  14.455 6.11e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8169 on 26 degrees of freedom
## Multiple R-squared:  0.8893, Adjusted R-squared:  0.8851
## F-statistic: 208.9 on 1 and 26 DF,  p-value: 6.111e-14
## [1] 3.783141
## [1] 2.509221
##      2.5%      97.5%
## 0.1619555 0.2095914
##      2.5%      97.5%
## 4.279862 3.307135
##      2.5%      97.5%
## 0.2398251 0.3167005
##      2.5%      97.5%
## 2.890219 2.188652
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```



#2.1

Italy Growth Rate: 0.183, deaths increased by 18.3% a day Italy Doubling Time: About 3.8 days to double
 95% Bootstrapped CI: We are 95% confident that the true growth rate is between [0.160, 0.209] We are 95% confident that the true doubling time is between [4.4, 3.4]

#2.2 Spain Growth Rate: 0.276, deaths increased by 27.6% a day Spain Doubling Time: About 2.5 days to double
 95% Bootstrapped CI: We are 95% confident that the true growth rate is between [0.232, 0.317] We are 95% confident that the true doubling time is between [3.0, 2.2]

```
## log_price
```

```
## -1.618578
```

```
##      2.5%      97.5%
```

```
## -1.772368 -1.446475
```

#3 $\log(\text{sales}) \sim \log(\text{price})$ was my equation. The estimated price elasticity demand is about -1.62, this means a 1% increase in milk price leads to approximately a 1.62% decrease in sales. This suggests that milk demand is elastic.

95% confidence interval of [-1.77, -1.45] elasticity estimate.